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# Software Engineering

## L AB 5: GIT AND GITHUB

dr. inz Abdulla Juwaied

# Learning Objectives

- What is Version Control
- Basic CLI Commands
- Introduction to Git
- Setting up VS Code
- Introduction to GitHub

# What is Version Control System?

A Version Control System manages the entire process of version and keeps track of the changes made.

- Track the changes made,
- Revert back to any versions when needed,
- Compare the changes made between any two versions,
- To know information like who, when, why made the changes,
- To get all these benefits with minimal effort.

# The "Undo" Button for Developers

- The Problem: Saving files as `project_final.zip`, `project_final_v2.zip`, and `project_FINAL_FOR_REAL.zip`.
- The Solution: Version Control.
- **Git**: The engine that tracks changes.
- **GitHub**: The cloud where those changes live and where teams collaborate.

# What is Git?



- Git is a **Distributed Version Control System**. It runs locally on your computer and keeps a detailed history of every tiny change you make to your code.
- Git saves the state of your project at specific points in time.
- Since it's local, it's lightning-fast.
- You can't lose data or corrupt your history easily.

# Setup Git



- Check version
- Git Add in your name
- Add in your email id
- Initialize an existing directory as a Git repo

```
$ git --version
```

```
$ git config --global user.name "[firstname lastname]"
```

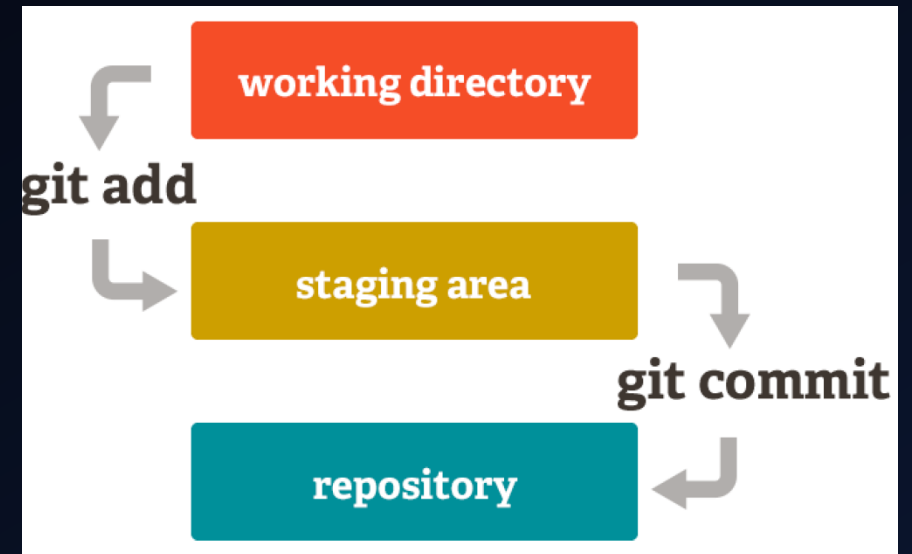
```
$ git config --global user.email "[valid-email]"
```

```
$ git init
```

# The Three Stages of Git



- **Working Directory:** Where you are currently typing and deleting code (Unmodified/Modified).
- **Staging Area (Index):** The "waiting room." You pick which changes are ready to be saved.
- **Repository (.git folder):** The permanent record. Once you "commit," the snapshot is stored forever.



# The "Big Four" commands you will use 90% of the time:

`git init`: Create a brand-new repository.

`git add <file>`: Move changes to the Staging Area.

`git commit -m "message"`: Save the snapshot to the Repository.

`git status`: Check which files are tracked or changed.



```
git status
```

```
git add [file]
```

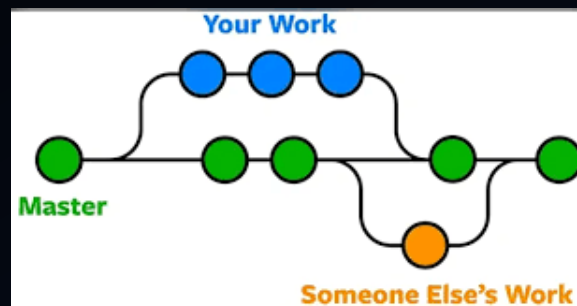
```
git reset [file]
```

```
$ git commit -m "Some valuable message"
```

```
$ git log
```



# Branching



```
$ git log
```

**It displays  
all the  
commits  
in the  
current  
branch.**

```
$ git branch
```

## List your branches

```
$ git branch [name]
```

**Creates a new branch  
from the current commit**

```
$ git checkout [name]
```

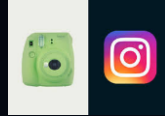
## Switches to another branch

```
$ git merge [branch]
```

**Merges the specified branch's history into the current one**

# What is GitHub?

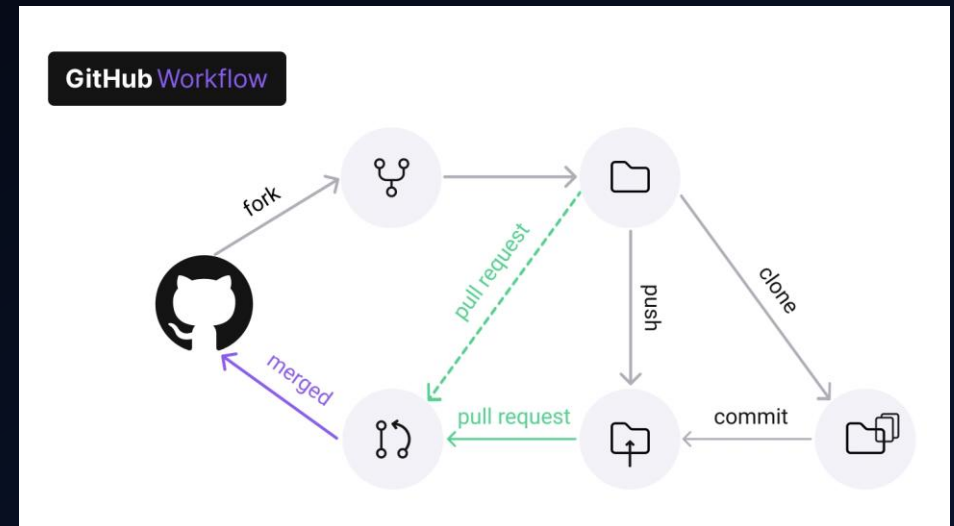
If Git is the "camera" taking snapshots, GitHub is the "Instagram" where you post them to share with the world.



- Till now, we have all the changes made saved in our local system only.
- To host our repository online, we need a server or hosting platform.
- GitHub is a repository hosting platform that uses Git in its core.
- Allows multiple people to work on the same code without overwriting each other.
- **Remember: Your GitHub profile is your "coding resume."**

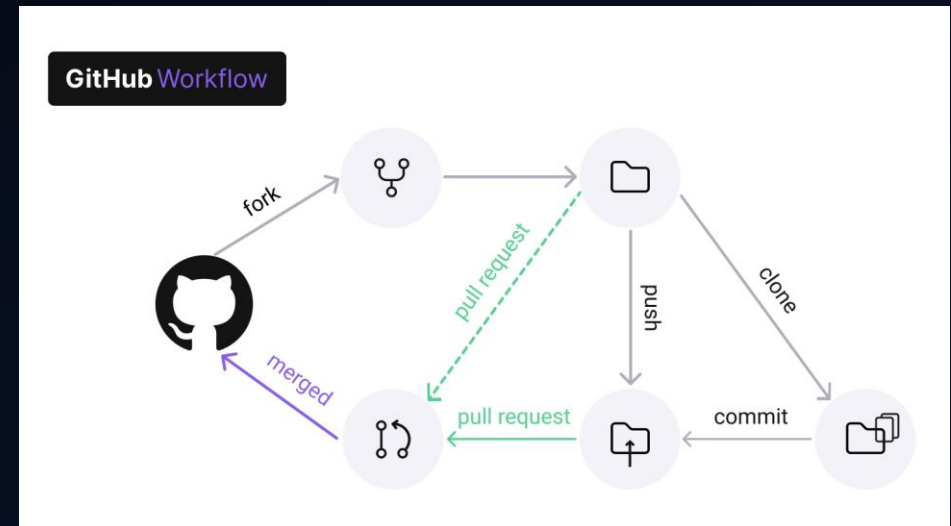
# The GitHub Workflow

- **Fork**: Copy someone else's project to your account.
- **Clone**: Download a repository to your local machine.
- **Push**: Send your local commits up to GitHub.
- **Pull**: Fetch and download the latest changes from GitHub to your machine.



## Tips for YOU

- Commit Often: Small commits are easier to fix than giant ones.
- Write Good Messages: "Fixed login bug" is better than "asdfghjkl."
- Don't Fear the Terminal: While there are desktop apps (GUIs), knowing the commands makes you a more versatile developer.
- The `.gitignore`: Use this file to tell Git which files (like passwords or huge data files) it should never track.



## Lab 5 Task: Publishing Your UML Portfolio

### Create a GitHub Account

- Go to [GitHub.com](https://github.com).
- Click **Sign up** in the top-right corner.
- Enter your **email address** and create a **strong password**.
- Choose a **Username**.



Create a main folder on your computer named Software-Engineering-Labs.  
Inside that folder, organise your existing work into subfolders:

/Lab2\_UseCase\_Diagrams

/Lab3\_Class\_Diagrams

/Lab4\_Behavioral\_Diagrams

### Create Your Online Repository

Log into your new GitHub account.

Click New Repository.

Repository Name: [SE\\_Labs\\_Portfolio](#)

### Connect and Push to GitHub

GitHub Integration Strategy SRS :

Option A: The Formal PDF/Word Document

Option B: The README.md

**Repository Owner Method** → One student creates the repository and adds teammates via **Settings** > **Collaborators**.